Commutative Algebra Problem Set #5

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Problem (Exercise 1. Variant of Cayley-Hamilton)**.** *Let M be a finitely generated A-module and let $\varphi : M \to M$ an A-module morphism. Let $\mathfrak{a} \subseteq A$ be an ideal such that $\varphi(M) \subseteq \mathfrak{a}M$. Show that φ satisfies an equation of the form*

$$\varphi^n + a_1\varphi^{n-1} + \cdots + a_n = 0$$

where $a_i \in \mathfrak{a}^i$ for $i \in [n]$.

Solution. Suppose $(e_1, \ldots, e_n) = M$ is a set of generators. Since $\varphi(M) \subseteq \mathfrak{a}M$:

$$\varphi(e_i) = \sum_{j=1}^{n} \gamma_{i,j} e_j, \quad \gamma_{i,j} \in \mathfrak{a}, \quad i \in [n].$$

This implies

$$\sum_{j=1}^{n} (\delta_{i,j}\varphi - \gamma_{i,j}) e_j = 0, \quad i \in [n],$$

and, in particular, that $\det\left(\varphi E_{n\times n} - \Gamma\right)$ for $\Gamma = (\gamma_{i,j})_{i,j\in[n]}$, $\gamma_{i,j} \in \mathfrak{a}$ is the zero endomorphism of M. When we have proven that $a_i \in \mathfrak{a}$, we simply stated that this determinant is clearly of form $\varphi^n + a_1\varphi^{n-1} + \cdots + a_n = 0$ for $a_i \in \mathfrak{a}$. Now we shall concretize that $a_i \in \mathfrak{a}^i$. For that, we shall show that each coefficient in the polynomial in φ is a sum of products of exactly i entries of Γ, hence lies in $\mathfrak{a}^i$.

This statement is more or less clear from linear algebra, but let us show this explicitly*. Denote by S_n the set of permutations $[n] \to [n]$ and by $\mathrm{Fix}(\sigma)$ the fixed points of $\sigma \in S_n$. Note that by definition of determinant:

$$\begin{aligned}
\det\left(\varphi E_{n\times n} - \Gamma\right) &= \sum_{\sigma\in S_n} \mathrm{sign}(\sigma) \prod_{i=1}^{n} (\delta_{i,\sigma(i)}\varphi - \gamma_{i,\sigma(i)}) \\
&= \sum_{\sigma\in S_n} (\pm 1) \prod_{i\in[n]\setminus \mathrm{Fix}(\sigma)} \gamma_{i,\sigma(i)} \prod_{i\in \mathrm{Fix}(\sigma)} (\varphi - \gamma_{i,i}) \\
&= \sum_{\sigma\in S_n} \beta_\sigma \prod_{i\in \mathrm{Fix}(\sigma)} (\varphi - \gamma_{i,i}),
\end{aligned}$$

where $\beta_\sigma = \pm \prod_{i\in[n]\setminus\mathrm{Fix}(\sigma)} \gamma_{i,\sigma(i)}$ (the exact value of sign is irrelevant for our discussion). Now, fix σ and suppose $t := \#\mathrm{Fix}(\sigma)$. Then, $\beta_\sigma \in \mathfrak{a}^{n-t}$. Now,

*What follows is a very rough brute-force which I am very sure might be greatly simplified. But I am too dumb to figure out some elegant solution

when expanding the product $\prod_{i\in \mathrm{Fix}(\sigma)}(\varphi - \gamma_{i,i})$, we notice that this expands into the sum $\sum_{j=0}^{t} \beta_j \varphi^j$ where clearly $\beta_j \in \mathfrak{a}^{t-j}$ (this coefficient is in fact exactly $\pm \sum_{|I|=t-j} \prod_{i\in I} \gamma_{i,i}$ where $I \subseteq \mathrm{Fix}(\sigma)$) and $\beta_t = 1$. Therefore,

$$\det\left(\varphi E_{n\times n} - \Gamma\right) = \sum_{\sigma\in S_n} \sum_{j=0}^{t} \underbrace{\beta_\sigma}_{\in \mathfrak{a}^{n-t}} \underbrace{\beta_j}_{\in \mathfrak{a}^{t-j}} \varphi^j,$$

so finally we conclude that for each $\sigma \in S_n$, the coefficient of φ^j comes from $\mathfrak{a}^{n-t+t-j} = \mathfrak{a}^{n-j}$. Note that the power of $\mathfrak{a}$ is independent of σ, and so after summing over all permutations we obtain the final result. □

Problem (Exercise 2). *Let $k \not\supseteq \mathbb{F}_2$ be a field and let A be a k-algebra. Construct a bijection between the sets of*

- ***involutions*** *on A, i.e. k-algebra homomorphisms $f : A \to A$ such that $f \circ f = id_A$.*
- ***$\mathbb{Z}/2$-gradings*** *on A, i.e. direct sum decompositions of the underlying abelian group $A \cong A_0 \oplus A_1$ such that $k \subseteq A_0$ and $A_i A_j \subseteq A_{(i+j) \bmod 2}$.*

Solution. Suppose $f : A \to A$ is an involution. Define

$$A_0 := \{a \in A : f(a) = a\}, \quad A_1 := \{a \in A : f(a) = -a\}.$$

Now note that any $a \in A$ can be written as (2 is invertible):

$$a = \frac{a + f(a)}{2} + \frac{a - f(a)}{2} \in A_0 \oplus A_1.$$

And also taking $x \in A_i$ and $y \in A_j$, we have

$$f(xy) = (-1)^i(-1)^j xy = (-1)^{i+j} xy \implies xy \in A_{(i+j) \bmod 2}.$$

Now suppose $A \cong A_0 \oplus A_1$ is a $\mathbb{Z}/2$-grading. Define:

$$f(a_0 + a_1) = a_0 - a_1, \quad a_0 \in A_0, a_1 \in A_1$$

Now check properties:

$$f^2(a_0, a_1) = f(a_0 - a_1) = a_0 + a_1 \implies f^2 = \mathrm{id}_A$$

and homomorphism:

$$\begin{aligned} f((a_0 + a_1)(b_0 + b_1)) &= f(a_0b_0 + a_0b_1 + a_1b_0 + a_1b_1) \\ &= a_0b_0 + a_1b_1 - (a_0b_1 + a_1b_0) \\ &= (a_0 - a_1)(b_0 - b_1) = f(a_0 + a_1)f(b_0 + b_1) \end{aligned}$$

Both maps are inverse and give a bijective correspondence.

Problem (Exercise 3)**.** *Prove that every UFD is normal.*

Solution. Suppose A is a UFD. We shall show that it is integral closed in its field of fractions: that is, if $x \in \mathrm{Frac}(A)$ is integral over A, then $x \in A$. Note that if $x = p/q$ is integral (and we may assume $\gcd(p, q) = 1$ since A is UFD), then

$$\left(\frac{p}{q}\right)^n + a_{n-1}\left(\frac{p}{q}\right)^{n-1} + \cdots + a_n = 0, \quad a_i \in A.$$

This implies

$$p^n + a_{n-1}qp^{n-1} + a_{n-2}q^2p^{n-2} + \cdots + a_nq^n = 0.$$

And this, in particular, implies $q \mid p^n$ or $q \mid p$ since they are co-prime. Therefore, $q \in A^\times$, so $p/q \in A$ and so A is integrally closed.

Problem (Exercise 4)**.** *A topological space X is called* ***Noetherian*** *if every increasing chain of open subsets stabilizes.*

(a) Let A be a Noetherian ring. Prove that $\mathrm{Spec}(A)$ is a Noetherian topological space.

(b) Does the converse hold?

(c) Prove that a topological space X is Noetherian iff every open subset of X is quasi-compact.

Solution.

Part (a). Recall that basis open sets of $\mathrm{Spec}(A)$ are $D(f) := \{\mathfrak{p} \not\supseteq f\}$. Suppose $U \subseteq \mathrm{Spec}(A)$ is some open subset and let $I(U) := \{f \in A : D(f) \subseteq U\}$. Trivially, $I(U) \subseteq A$ is an ideal. Additionally, $D(I(U)) = U$.

Now take some increasing sequence of open sets:

$$U_1 \subseteq U_2 \subseteq U_3 \subseteq \dots$$

The corresponding ideals satisfy:

$$I(U_1) \subseteq I(U_2) \subseteq I(U_3) \subseteq \dots$$

Since A is Noetherian, this chain stabilizes: say, $I(U_n) = I(U_N)$ for all $N \geq n$. But then $D(I(U_n)) = D(I(U_N))$ or $U_n = U_N$ for all $N \geq n$. □

Part (b). Take $A := k[x_1, x_2, \dots]/(x_1^2, x_2^2, \dots)$. This ring is clearly non-Noetherian (take chain of ideals $(x_1) \subseteq (x_1, x_2) \subseteq (x_1, x_2, x_3) \subseteq \dots$). Denote $\mathfrak{m} := (x_1, x_2, \dots)$.

Notice that $\sqrt{0} = \mathfrak{m}$. Further note that $A/\mathfrak{m} \cong k$, so $\mathfrak{m}$ is the unique prime ideal of A, thus $\mathrm{Spec}(A) = \{\mathfrak{m}\}$, and so it is trivially Noetherian.

Part (c). qc⇒Noetherian. This direction is easier: suppose $U_1 \subseteq U_2 \subseteq U_3 \subseteq \dots$ is a chain of open sets and let $U := \bigcup_{n\in\mathbb{N}} U_n$. Note that U is open and $\{U_n\}_{n\in\mathbb{N}}$ is its cover. Due to quasi-compactness, finitely many of U_n cover U, implying that the chain stabilizes.

Noetherian⇒qc. Let X be Noetherian and $U \subseteq X$ open. Let $\{U_\alpha\}_{\alpha\in I}$ be an open cover of U. Define family $\mathcal{V} := \{V \subseteq U : V \text{ is covered by finitely many } U_\alpha\}$, ordered by inclusion. Now note that we can apply Zorn's lemma to find the maximal element. Why? Note that every chain $V_1 \subseteq V_2 \subseteq V_3 \subseteq \dots$ stabilizes due to X being Noetherian, and so the upper bound is some $V_N \in \mathcal{V}$. Let the maximal element be W. We shall prove that $W = U$. Assume, however, otherwise. Then, we can pick $x \in U \setminus W$. Choose some U_j such that $x \in U_j$. Then, $W \cup U_j$ is a finite union of U_α's and strictly contains W, so W is not maximal. Contradiction. Thus, U is a finite union of U_α's, so every open U is quasi-compact. □

Problem (Exercise 5)**.** *Let B be a ring and M a B-module. We make the direct sum $B \oplus M\varepsilon$ into a ring with multiplication*

$$(b + m\varepsilon)(b' + m'\varepsilon) = bb' + (bm' + b'm)\varepsilon$$

The subgroup $I = 0 \oplus M\varepsilon$ is an ideal with $I^2 = 0$ and quotient $(B \oplus M\varepsilon)/I \cong B$. We denote by $\pi : B \oplus M\varepsilon \to B$ the quotient map.
Let $A \to B$ be a map of rings and M a B-module. Construct the bijection between the sets of

- *A-linear derivations $\delta : B \to M$.*
- *A-algebra homomorphisms $\varphi : B \to B \oplus M\varepsilon$ s.t. $\pi \circ \varphi = id_B$.*

Solution. Interestingly, this problem seems very related to how one works with infinitesimals in Physics. For instance, suppose we want to linearize some function $f : \mathbb{R} \to \mathbb{R}$ near b. Then, we have $f(b + \varepsilon) \approx f(b) + f'(b)\varepsilon$ for some very small ε. Here, by "small" we concretely mean $\varepsilon^2 = 0$ (so we neglect all quadratic terms in the expansion). Then, if we wish to multiply two functions, linearized by this relation, we get:

$$\begin{aligned} f(b+\varepsilon)g(b+\varepsilon) &= (f(b) + f'(b)\varepsilon)(g(b) + g'(b)\varepsilon) \\ &\approx f(b)g(b) + (f(b)g'(b) + f'(b)g(b))\varepsilon \\ &\approx f(b)g(b) + (fg)'(b)\varepsilon \end{aligned}$$

by neglecting the ε^2 term.

So this sort of motivates us to, based on derivation $\delta : B \to M$, define the B-algebra homomorphism as $\varphi_\delta = b + \delta(b)\varepsilon$. Trivially, $\pi \circ \varphi_\delta = \mathrm{id}_B$. Also, additivity and A-linearity are obvious. Check multiplicativity:

$$\begin{aligned}\varphi_\delta(b)\varphi_\delta(b') &= (b + \delta(b)\varepsilon)(b' + \delta(b')\varepsilon) \\ &= bb' + (b\delta(b') + b'\delta(b)) \\ &= bb' + \delta(bb') = \varphi_\delta(bb')\end{aligned}$$

Now compose the reverse direction. Suppose we are given $\varphi : B \to B \oplus M\varepsilon$ with $\pi \circ \varphi = \mathrm{id}_B$. Since composition gives identity, $\varphi(b) = b + \delta_\varphi(b)\varepsilon$. Define $\delta_\varphi : B \to M$ to be the second component of φ as above. Additivity and A-linearity are again clear, so let us check whether this gives A-derivation. Notice that since φ is a homomorphism:

$$\begin{aligned}\varphi(bb') = \varphi(b)\varphi(b') &= (b + \delta_\varphi(b)\varepsilon)(b' + \delta_\varphi(b')\varepsilon) \\ &= bb' + (\delta_\varphi(b)b' + \delta_\varphi(b')b),\end{aligned}$$

so $\delta(bb') = \delta_\varphi(b)b' + \delta_\varphi(b')b$, as required.

Finally, evidently $\delta_{\varphi_\delta} = \delta$ and $\varphi_{\delta_\varphi} = \varphi$, so we indeed have a bijection. □